Naoyuki Kamiyama

Curriculum Vitæ

Position

Apr. 2026	–		Professor, Kyoto University Graduate School of Informatics Applied Mathematics and Physics Course
Jul. 2019	–	Mar. 2026	Professor, Kyushu University Institute of Mathematics for Industry
Oct. 2011	–	Jun. 2019	Associate Professor, Kyushu University Institute of Mathematics for Industry
Apr. 2009	–	Sep. 2011	Assistant Professor, Chuo University Faculty of Science and Engineering Department of Information and System Engineering
Concurrent position			
Oct. 2014	–	Mar. 2021	PRESTO Researcher, JST, PRESTO
Apr. 2007	–	Mar. 2009	Research Fellow of the Japan Society for the Promotion Science

Education

Mar. 2009	Doctor of Engineering, Kyoto University Graduate School of Engineering Department of Architecture and Architectural Engineering
Mar. 2006	Master of Engineering, Kyoto University Graduate School of Engineering Department of Architecture and Architectural Engineering
Mar. 2004	Bachelor of Engineering, Kyoto University Faculty of Engineering Undergraduate School of Architecture

Peer-Reviewed Journal Papers

78. Naoyuki Kamiyama
The Set of Stable Matchings and the Core in a Matching Market with Ties and Matroid Constraints
International Journal of Game Theory, accepted.

77. Takasugu Shigenobu and Naoyuki Kamiyama
A Necessary Condition for Connectedness of Solutions to Integer Linear Systems
Journal of the Operations Research Society of Japan, accepted.

76. Naoyuki Kamiyama
The Strongly Stable Roommates Problem and Linear Programming
Discrete Optimization, vol.62, pp.100969 (2026).

75. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yoshio Okamoto
Loss Minimization for Electrical Flows over Spanning Trees on Grids
Theoretical Computer Science, vol.1083, pp.116132 (2026).

74. Yuen Sheung Man, Ayumi Igarashi, Naoyuki Kamiyama, Warut Suksompong
Reforming an Unfair Allocation by Exchanging Goods
Algorithmica, vol.88, no.4, pp.57 (2026).

73. Naoyuki Kamiyama
The Strongly Stable Matching Problem with Closures
Theoretical Computer Science, vol.1076, pp.115980 (2026).

72. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Shun-ichi Maezawa, Yuta Nozaki, Yoshio Okamoto
Hardness of Finding Combinatorial Shortest Paths on Graph Associahedra
SIAM Journal on Discrete Mathematics, vol.40, no.2, pp.554–575 (2026).

71. Naoyuki Kamiyama
Non-uniformly Stable Matchings
ACM Transactions on Economics and Computation, vol.14, no.1, pp.1:1-1:20 (2026).

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69. Takehiro Ito, Yuni Iwamasa, Naoyuki Kamiyama, Yasuaki Kobayashi, Yusuke Kobayashi, Shun-ichi Maezawa, Akira Suzuki
Reconfiguration of Time-Respecting Arborescences
Algorithmica, vol.88, no.1, pp.15 (2026).

68. Naoyuki Kamiyama
Strongly Stable Matchings under Matroid Constraints
SIAM Journal on Discrete Mathematics, vol.39, no.3, pp.1471–1490 (2025).

67. Naoyuki Kamiyama
The Minimum-Cost Dynamic Flow Problem in a Fixed Graph with a Constant Target Flow Value
Networks, vol.85, no.4, pp.437–444 (2025).

66. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yoshio Okamoto
Algorithmic Theory of Qubit Routing in the Linear Nearest Neighborhood Architectures
ACM Transactions on Quantum Computing, vol.6, no.3, pp.19:1–19:17 (2025).

65. Naoyuki Kamiyama
Super-Stable Common Independent Sets of Generalized Matroids
Operations Research Letters, vol.60, pp.107248 (2025).

64. Naoyuki Kamiyama
Modifying an Instance of the Super-Stable Matching Problem
Information Processing Letters, vol.189, pp.106549 (2025).

63. Takehiro Ito, Yuni Iwamasa, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yuta Nozaki, Yoshio Okamoto, Kenta Ozeki
Reforming an Envy-Free Matching
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62. Ayumi Igarashi, Naoyuki Kamiyama, Warut Suksompong, Yuen Sheung Man
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61. Kristóf Bérczi, Erika Bérczi-Kovács, Endre Boros, Fekadu Tolessa Gedefa, Naoyuki Kamiyama, Telikepalli Kavitha, Yusuke Kobayashi, Kazuhisa Makino
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60. Naoyuki Kamiyama
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59. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yuta Nozaki, Yoshio Okamoto, Kenta Ozeki
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58. Naoyuki Kamiyama
On Optimization Problems in Acyclic Hypergraphs
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57. Masataka Shirahashi and Naoyuki Kamiyama
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56. Naoyuki Kamiyama
Pareto Efficient Matchings with Pairwise Preferences
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54. Naoyuki Kamiyama
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49. Naoyuki Kamiyama
Envy-Free Matchings with One-Sided Preferences and Matroid Constraints
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45. Naoyuki Kamiyama
The Distance-Constrained Matroid Median Problem
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Exact and Approximation Algorithms for Weighted Matroid Intersection
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38. Susumu Kawanaka and Naoyuki Kamiyama
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Journal of Combinatorial Optimization, vol.36, no.10, pp.1299–1314 (2018).

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A Characterization of Weighted Popular Matchings under Matroid Constraints
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Efficient Stabilization of Cooperative Matching Games
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Strategic Issues in College Admissions with Score-Limits
Operations Research Letters, vol.45, pp.105–108 (2017).

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Theoretical Computer Science, vol.659, pp.95–97 (2017).

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27. Naoyuki Kamiyama
The Popular Matching and Condensation Problems under Matroid Constraints
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26. Satoru Iwata, Naoyuki Kamiyama, Naoki Katoh, Shuji Kijima, Yoshio Okamoto
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25. Keita Nakamura and Naoyuki Kamiyama
Many-to-Many Stable Matchings with Ties in Trees
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23. Naoyuki Kamiyama and Yasushi Kawase
On Packing Arborescences in Temporal Networks
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The Nucleolus of Arborescence Games in Directed Acyclic Graphs
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21. Naoyuki Kamiyama and Naoki Katoh
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Discrete Applied Mathematics, vol.178, pp.89–100 (2014).

20. Yuya Higashikawa, Naoyuki Kamiyama, Naoki Katoh, Yuki Kobayashi
An Inductive Construction of Minimally Rigid Body-Hinge Simple Graphs
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19. Naoyuki Kamiyama
A New Approach to the Pareto Stable Matching Problem
Mathematics of Operations Research, vol.39, no.3, pp.851–862 (2014).

18. Naoyuki Kamiyama
Arborescence Problems Theorems and Algorithms
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17. Yusuke Matsumoto, Naoyuki Kamiyama, Keiko Imai
On Total Unimodularity of Edge-Edge Adjacency Matrices
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16. Naoyuki Kamiyama
A Note on the Serial Dictatorship with Project Closures
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15. Naoyuki Kamiyama
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14. András Frank, Satoru Fujishige, Naoyuki Kamiyama, Naoki Katoh
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13. Naoyuki Kamiyama
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12. Naoyuki Kamiyama
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11. Satoru Fujishige and Naoyuki Kamiyama
The Root Location Problem for Arc-Disjoint Arborescences
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10. Yuichiro Yasui, Katsuki Fujisawa, Kazushige Goto, Naoyuki Kamiyama, Mizuyo Takamatsu
NETAL High-performance Implementation of Network Analysis Library Considering Computer Memory Hierarchy
Journal of the Operations Research Society of Japan, vol.54, no.4, pp.259–280 (2011).

9. Yusuke Matsumoto, Naoyuki Kamiyama, Keiko Imai
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8. Yutaka Iwaikawa, Naoyuki Kamiyama, Tomomi Matsui
Improved Approximation Algorithms for Firefighter Problem on Trees
IEICE Transaction on Information and Systems, vol.E94-D, no.2, pp.196–199 (2011).

7. Naoyuki Kamiyama and Naoki Katoh
Covering Directed Graphs by In-Trees
Journal of Combinatorial Optimization, vol.21, no.1, pp.2–18 (2011).

6. Kristóf Bérczi, Satoru Fujishige, Naoyuki Kamiyama
A Linear-Time Algorithm to Find a Pair of Arc-Disjoint Spanning In-Arborescence and Out-Arborescence in a Directed Acyclic Graph
Information Processing Letters, vol.109, no.23–24, pp.1227–1231 (2009).

5. Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
An Efficient Algorithm for the Evacuation Problem in a Certain Class of Networks with Uniform Path-Lengths
Discrete Applied Mathematics, vol.157, no.17, pp.3665–3677 (2009).

4. Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
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Combinatorica, vol.29, no.2, pp.197–214 (2009).

3. Naoyuki Kamiyama, Yuuki Kiyonari, Eiji Miyano, Shuichi Miyazaki, Katsuhisa Yamanaka
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IEICE Transaction on Information and Systems, vol.E92-D, no.2, pp.130–140 (2009).

2. Takenao Taji, Shin-ichi Tanigawa, Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
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1. Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
An Efficient Algorithm for Evacuation Problem in Dynamic Network Flows with Uniform Arc Capacity
IEICE Transaction on Information and Systems, vol.E89-D, no.8, pp.2372–2379 (2006).

Peer-Reviewed Conference Papers

(Especially, in SODA, AAMAS, AAAI, IJCAI, WSC, LNCS, LIPIcs, CRPIT, CCIS)

46. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yoshio Okamoto
Vertex Cover Interdiction in Bipartite Graphs
37th International Symposium on Algorithms and Computation (ISAAC), accepted.

45. Yasunari Hikima, Naoyuki Kamiyama, Shinsaku Sakaue, Taira Tsuchiya
A Sampling-Based Approach to Contextual Inverse Optimization
Proceedings of the 35th International Joint Conference on Artificial Intelligence (IJCAI-ECAI), accepted.

44. Ayumi Igarashi, Naoyuki Kamiyama, Yasushi Kawase, Warut Suksompong, Hanna Sumita, Yu Yokoi
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Proceedings of the 21st Conference on Web and Internet Economics (WINE), LNCS 16266, pp.782 (2026).

43. Naoyuki Kamiyama, Hiroaki Yamada, Takashi Kato, Shizuo Kaji, Tetsuro Takahashi
A Topological Data Analysis Approach to Detecting Congestion in Pedestrian Crowds
Proceedings of the 2025 Winter Simulation Conference (WSC), 2670-2681 (2025).

42. Yuen Sheung Man, Ayumi Igarashi, Naoyuki Kamiyama, Warut Suksompong
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Proceedings of the 36th International Symposium on Algorithms and Computation (ISAAC), LIPIcs 359, pp.54:1–54:21 (2025).

41. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yoshio Okamoto
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Proceedings of the 36th International Symposium on Algorithms and Computation (ISAAC), LIPIcs 359, pp.40:1–40:14 (2025).

40. Yasunari Hikima and Naoyuki Kamiyama
An Inverse Optimization Approach to Contextual Inverse Optimization

Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), 5354–5362 (2025).

39. Ayumi Igarashi, Naoyuki Kamiyama, Warut Suksompong, Yuen Sheung Man
Reachability of Fair Allocations via Sequential Exchanges
Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI), pp.9773–9780 (2024).

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Hardness of Finding Combinatorial Shortest Paths on Graph Associahedra
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On Reachable Assignments under Dichotomous Preferences
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32. Hiroaki Yamada, Masataka Shirahashi, Naoyuki Kamiyama, Yumeka Nakajima
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Monotone Edge Flips to an Orientation of Maximum Edge-Connectivity à la Nash-Williams
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Proceedings of the 23rd International Conference on Principles and Practice of Multi-Agent Systems (PRIMA), LNCS 12568, pp.299–314 (2021).

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21. Hiroaki Yamada, Kotaro Ohori, Tadashige Iwao, Akifumi Kira, Naoyuki Kamiyama, Hiroaki Yoshida, Hirokazu Anai
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20. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yoshio Okamoto
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17. Chien-Chung Huang, Naonori Kakimura, Naoyuki Kamiyama
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Proceedings of the 27th Annual ACM/SIAM Symposium on Discrete Algorithms (SODA), pp.430–444 (2016).

16. Naoyuki Kamiyama
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Proceedings of the 8th International Symposium on Algorithmic Game Theory (SAGT), LNCS 9347, pp.3–14 (2015).

15. Takehiro Ito, Naonori Kakimura, Naoyuki Kamiyama, Yusuke Kobayashi, Yoshio Okamoto
Minimum-Cost b-Edge Dominating Sets on Trees
Proceedings of the 25th International Symposium on Algorithms and Computation (ISAAC), LNCS 8889, pp.195–207 (2014).

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Proceedings of the 23rd Annual ACM/SIAM Symposium on Discrete Algorithms (SODA), pp.135–142 (2012).

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Robustness of Minimum Cost Arborescences
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9. Naoyuki Kamiyama
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8. Naoyuki Kamiyama and Tomomi Matsui
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7. Naoyuki Kamiyama
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6. Naoyuki Kamiyama and Naoki Katoh
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5. Naoyuki Kamiyama and Naoki Katoh
The Minimum Weight In-Tree Cover Problem
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4. Naoyuki Kamiyama and Naoki Katoh
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Proceedings of the 14th Annual International Computing and Combinatorics Conference (COCOON), LNCS 5092, pp.444–457 (2008).

3. Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
Arc-Disjoint In-Trees in Directed Graphs
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2. Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
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1. Naoyuki Kamiyama, Naoki Katoh, Atsushi Takizawa
An Efficient Algorithm for Evacuation Problems in Dynamic Network Flows with Uniform Arc Capacity
Proceedings of the 2nd International Conference on Algorithmic Aspect in Information and Management (AAIM), LNCS 4041, pp.231–242 (2006).

Scientific Service

Journal Editor

International Journal of Mathematics for Industry, subject editor (2023–)

Peer-Reviewed Conference Program Committee (Algorithm)

8th International Symposium on Combinatorial Optimization (ISCO), 2024
13th International Conference on Algorithms and Complexity (CIAC), 2023
7th International Symposium on Combinatorial Optimization (ISCO), 2022
31st International Symposium on Algorithms and Computation (ISAAC), 2020
14th International Frontiers of Algorithmics Workshop (FAW), 2020
6th International Symposium on Combinatorial Optimization (ISCO), 2020
5th International Symposium on Combinatorial Optimization (ISCO), 2017
28th International Symposium on Algorithms and Computation (ISAAC), 2017
4th International Symposium on Combinatorial Optimization (ISCO), 2016

Peer-Reviewed Conference Program Committee (Artificial Intelligence)

41st AAAI Conference on Artificial Intelligence (AAAI), 2027
35th International Joint Conference on Artificial Intelligence (IJCAI-ECAI), 2026
40th AAAI Conference on Artificial Intelligence (AAAI), 2026
28th European Conference on Artificial Intelligence (ECAI), 2025
39th AAAI Conference on Artificial Intelligence (AAAI), 2025
33rd International Joint Conference on Artificial Intelligence (IJCAI), 2024
38th AAAI Conference on Artificial Intelligence (AAAI), 2024
32nd International Joint Conference on Artificial Intelligence (IJCAI), 2023
37th AAAI Conference on Artificial Intelligence (AAAI), 2023
36th AAAI Conference on Artificial Intelligence (AAAI), 2022
35th AAAI Conference on Artificial Intelligence (AAAI), 2021
34th AAAI Conference on Artificial Intelligence (AAAI), 2020

Peer-Reviewed Conference Organizing Committee

32nd International Symposium on Algorithms and Computation (ISAAC), chair, 2021

Other Notable Conference Service

8th International Workshop on Matching Under Preferences (MATCHU-UP), PC, 2026
7th International Workshop on Matching Under Preferences (MATCHU-UP), PC, 2024
11th Hungarian-Japanese Symposium on Discrete Mathematics and Its Applications (HJ), PC, 2019
5th International Workshop on Matching Under Preferences (MATCHU-UP), PC, 2019
9th Hungarian-Japanese Symposium on Discrete Mathematics and Its Applications (HJ), PC chair, 2015
9th Hungarian-Japanese Symposium on Discrete Mathematics and Its Applications (HJ), OC, 2015

Journal Referee

ACM Transactions on Algorithms, ACM Transactions on Economics and Computation, Algorithmica, Artificial Intelligence, Autonomous Agents and Multi-Agent Systems Combinatorica, Discrete Applied Mathematics, Discrete Mathematics, Discrete Optimization, EURO Journal on Computational Optimization, European Journal of Combinatorics Games and Economic Behavior, Graphs and Combinatorics, IEICE Transactions, Information Processing Letters, International Journal of Mathematics for Industry, Japan Journal of Industrial and Applied Mathematics, Journal of Combinatorial Optimization, Journal of the Operations Research Society of Japan, Journal of Mathematical Economics, Mathematical Programming, Mathematics of Operations Research, Networks, Optimization Methods and Software, Optimization Letters, Pacific Journal of Optimization, RAIRO – Operations Research, RIMS Kokyuroku Bessatsu, SIAM Journal on Discrete Mathematics, SIAM Journal on Optimization, The International Journal of Foundation of Computer Science, and Theoretical Computer Science.

Conference Referee

AAAI Conference on Artificial Intelligence (AAAI), Annual International Conference on Combinatorial Optimization and Applications (COCOA), Annual International Computing and Combinatorics Conference (COCOON), Annual European Symposium on Algorithms (ESA), International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX), International Frontiers of Algorithmics Workshop (FAW), International Symposium on Fundamentals of Computation Theory (FCT), Annual Symposium on Foundations of Computer Science (FOCS), IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS), International Colloquium on Automata, Languages, and Programming (ICALP), International Conference on Integer Programming and Combinatorial Optimization (IPCO), International Conference on Autonomous Agents and MultiAgent Systems (AAMAS), International Conference on Algorithms and Complexity (CIAC), International Conference on Language and Automata Theory and Applications (LATA), International Joint Conference on Artificial Intelligence (IJCAI) International Symposium on Algorithms and Computation (ISAAC), International Symposium on Algorithmic Game Theory (SAGT), International Symposium on Combinatorial Optimization (ISCO) International Symposium on Scheduling (ISS), International Workshop on Computational Social Choice (COMSOC), Latin American Theoretical Informatics Symposium (LATIN), International Symposium on Mathematical Foundations of Computer Science (MFCS), International Conference on Principles of Distributed Systems (OPODIS), Annual ACM/SIAM Symposium on Discrete Algorithms (SODA), Symposium on Theoretical Aspects of Computer Science (STACS), Algorithms and Data Structures Symposium (WADS), Workshop on Algorithms and Computation (WALCOM), and International Workshop on Graph-Theoretic Concepts in Computer Science (WG).

Ph.D. Student

Takashi Ishizuka (Kyushu University, 2022)